SIMATS ENGINEERING
ASSESSMENT TOOL 1: Short Answer Test

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions.* (BL2)

Total Marks: 25

Dave's Taxonomy: Imitation (Level 1)
Question Count: 5

Code of Conduct

1 K.V. yathish chandra (192524323) (Name / Reg. No.)

certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: _____

Assessment Tasks
Short Answer Test

1. Define cloud computing and explain any four characteristics that make it different from traditional on-premise computing.
2. Trace the evolution of cloud computing from hardware evolution to internet software evolution, and show how these changes enabled modern cloud platforms.
3. Explain the role of server virtualization in cloud computing. Differentiate between virtualization as an *enabling technology* and *cloud computing as a service model.*
4. Discuss the role of **data centers** in delivering cloud services..
5. **Differentiate IaaS, PaaS, SaaS, and XaaS with one suitable real-world example for each model.**

Short Answer Test Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Conceptual Accuracy	30%	Accurate definitions and precise explanation of all key terms	Mostly accurate with minor gaps	Partial understanding with noticeable errors	Incorrect or irrelevant explanation
Coverage of Concepts	20%	Covers all expected sub-points completely	Covers most expected points	Covers only basic points	Very limited coverage
Use of Examples	15%	Uses highly relevant cloud examples to strengthen the answer	Uses relevant examples with minor mismatch	Uses vague or partially relevant examples	No useful examples
Comparison / Differentiation	20%	Clearly distinguishes concepts with strong logic	Reasonable differentiation with minor overlap	Comparison is weak or incomplete	Fails to differentiate concepts
Clarity of Expression	15%	Clear, organized, and concise answer writing	Generally clear with few issues	Understandable but poorly organized	Difficult to follow

Total Marks:

K.V.Yathish chandra-
192521323
cloud computing

① Define cloud computing and explain any four characteristics that make it different from traditional on-premise computing.

cloud computing is the delivery of computing resources such as servers, storage, databases, software, and networking over the Internet users can access these resource whenever requried without owning and maintaining the physical infrastructure

Four characteristics

(i) on-demand self service

users can create or use computing resources whenever needed without direct assistance from the service provider

(ii) Broad network access

cloud services can be accesed through the Internet using devices such as laptops, smartphones, and tabets

(iii) Resource Pooling

cloud providers share computing resources among multiple users using virtulization while maintaining proper security and isolation

Rapid elasticity:
Resources can be quickly increased or dec
sed according to demand. For example an online
shopping website can and servers during a
-festival sale

Differences from traditional computing
In traditional on-premise computing
an organization must purchase, install, and
maintain its own hardware and software

② Trace the evolution of cloud computing from
hardware evolution to Internet software
evolution, and show how these changes enabled
modern cloud platforms
cloud computing developed through several
important stages:
mainframe computing → client-server → compute
ing → virtualization → Internet/web services →
cloud computing.

i) mainframe computing
In the early stage, many users shared a
powerful central mainframe computer through
terminals

ii) client-server computing
computing responsibilities were divided
b/w clients and servers this allowed organizations
to provide applications and data through networke
d computers

(iii) Virtualization:
virtualization allowed one physical server to run multiple virtual machines. This improved hardware utilization and reduce infrastructure

(iv) Internet and web services
the growth of the Internet of web technologies made it possible to access applications and data - from anywhere

(v) Modern cloud platforms
cloud provides combined virtualization, high speed networks, distributed data centers, automation and web services

③ Explain the role of server virtualization in cloud computing. Differentiate b/w virtualization as an enabling technology and cloud computing as a service model

Role of server virtualization
Server virtualization divides one a physical server into multiple virtual machines (VMs) using a software layer called a hypervisor

For example
Physical server → hypervisor → VM1 + VM2 + VM3
Each VM can run its own operating system and applications

Benefits

* Improve physical server utilization
* Reduces hardware and electricity costs
* Allows multiple applications to share one server
* Makes resource allocation flexible
* Supports easy scaling and VM migration

Virtualization

* Technology used to create virtual resources
* Uses hypervisors and virtual machines
* Mainly focuses on resources abstraction
* Ex: VMware virtual machine

cloud computing

* Service model for delivering computing resources
* Uses data centers, networks, virtualization and automation
* Focuses on providing resources as services
* Ex: Amazon EC2

Conclusion:-

Virtualization is an important enabling technology, while cloud computing uses virtualization and other technologies to deliver computing services over the Internet

Roles of data centers

(i) computing resources

Data centers contain powerfull servers that run cloud applications and virtual machines

(ii) Data Storage

They store user-file, databases, backups and application data

(iii) Networking

High-speed networks connect servers and provide Internet access to users

(iv) availability and reliability

Backup power, redundant hardware, and multiple network connections help keep services available

(v) Scalability

cloud providers can add more servers and storage when user demand increases

Differentiable (aas paas, saas and xaas (with one suitable real world example for each model

Model	meaning	wait it provides	Example
Iaas	Infrastructure as a service	virtual machines	Amazon EC2
Paas	Platform as a service	Development Platforms	Google APP Engine
saas	Software as a service	Read to use software through Internet	Google Docs
Xaas	Anything as a service	Different IT resources delivered as a service	Amazon RD

Iaast
The provider given basic infrastructure such as virtual servers, storage, and networking. The customer manages the operating system and applications

Paast
The provider gives a ready-made platforms for developing and deploying applications. The developer mainly focuses on application development

Saast
The complete software application is provided through the Internet users simply access and use the application

Xaast
"Anything as a service" is a board concept where different IT capabilites are provided as services on demand

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